Sushant Bhattarai

Graduate Research Assistant / PhD Student, Computer Science
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RESEARCH INTERESTS

Distributed algorithms; learning-augmented algorithms; online algorithms; routing and repairperson problems; distributed directories; graph analytics; data engineering systems.

EDUCATION

Kent State University, Kent, OH, USA Aug 2024 – Present
Ph.D. in Computer Science (August 2024 – Present)
GPA: 4.0

MS in Computer Science (August 2024 – December 2026)
GPA: 4.0
Master's Thesis: *Online Learning-Augmented Distributed Directories*

Tribhuvan University, Kathmandu, Bagmati, Nepal Nov 2017 – Apr 2022
Bachelor's of Engineering (BE) in Computer Engineering
GPA: 80.81%
Undergraduate Thesis: *English Sign Gesture Recognition using CNN*

PUBLICATIONS

- **S. Bhattarai**, B. Maharjan, G. Sharma, and S. Guragain, "Online Learning-Augmented Distributed Directories," *2026 28th International Symposium on Stabilization, Safety, and Security of Distributed Systems (SSS)*, Gothenburg, Sweden, October 2026 (Accepted, to be published).
- S. Guragain, B. Maharjan, **S. Bhattarai**, G. Sharma, and P. Poudel, "Learning-Augmented Distributed Directories*," *2025 23rd International Symposium on Network Computing and Applications (NCA)*, Lisbon, Portugal, 2025, pp. 25–34, doi: 10.1109/NCA67271.2025.00015.
- **S. Bhattarai**, Y. Bhurtel, B. KC, and S. Subedi, "English Sign Gesture Recognition using CNN," *Proceedings of the 12th IOE Graduate Conference*, Kathmandu, Nepal, 2022, pp. 1825–1831.

RESEARCH EXPERIENCE

Graduate Research Assistant, Department of Computer Science Aug 2024 – Present
Kent State University, Kent, OH, USA
Advisor: Dr. Gokarna Sharma

- Conducting research on learning-augmented distributed directory protocols for shared-object access in large-scale distributed systems.
- Designing tree-based and cluster-based protocols using error-prone request predictions to reduce communication cost in distributed object access.
- Performing literature reviews on distributed algorithms, online algorithms, learning-augmented algorithms, and systems research.

Current Research Project: Multi-Repairperson Online TRP on a Line Ongoing

- Studying the online traveling repairperson problem on a line with multiple repairpersons and requests released over time.
- Developing and analyzing routing-style online optimization algorithms where each repairperson only knows requests released so far.

Undergraduate Thesis: English Sign Gesture Recognition using CNN May 2021 – Apr 2022
Tribhuvan University, Kathmandu, Nepal

- Designed and trained a CNN model to recognize English sign-language hand gesture alphabets from A to Z.
- Used Python, OpenCV, and TensorFlow for image processing, model training, and gesture classification.

- Published the work and presented a poster at the 12th Institute of Engineering Graduate Conference, Nepal, in October 2022.
- Code: github.com/sushant-bhattarai/sign-language-recognition-major-project

TEACHING EXPERIENCE

Assistant Lecturer May 2022 – Jul 2024
Himalaya College of Engineering, Tribhuvan University, Kathmandu, Nepal

- Lectured undergraduate students in Computer, Electronics, Communication, and Information Engineering.
- Courses lectured: Computer Graphics, Numerical Methods, and Introduction to Information Technology.
- Conducted related laboratory sessions and assisted fellow lecturers during laboratory periods.

Part-time Lecturer Nov 2022 – Jul 2024
Sagarmatha Engineering College, Tribhuvan University, Kathmandu, Nepal

- Course Lectured: Computer Graphics to undergraduate Electronics, Communication, and Information Engineering students.
- Conducted laboratory tasks related to the course.

SELECTED ACADEMIC PROJECTS

Distributed Subgraph Matching with Neighborhood-Signature Pruning Big Data Analytics

- Developed a distributed subgraph matching framework using Hadoop MapReduce, Hadoop Streaming, HDFS, Java, and Python.
- Implemented k-hop neighborhood-signature pruning to remove structurally incompatible candidates before verification.
- Designed a three-phase distributed pipeline for signature construction, candidate generation, and subgraph verification.
- Code: github.com/sushantbhattaraii/neighbor-signature-subgraph-matching

CarViz: Multi-View Coordinated Visualization System for Car Data Graduate Information Visualization

- Built a browser-based visualization system for exploratory analysis of a 406-record Cars Dataset from 1970–1982.
- Implemented coordinated multiple views in D3.js v7, including a brushable scatter plot, temporal line chart, grouped bar chart, and box-and-whisker plot.
- Code: github.com/sushantbhattaraii/info-viz-multiview-project

High Schools Recommendation System using Collaborative Filtering Sep 2020 – Mar 2021

- Built a Laravel-based web application that recommends high schools to users based on their interests.
- Implemented user-facing functionality allowing users to apply to recommended schools.
- Code: github.com/sushant-bhattarai/commoncollegeapplication

TECHNICAL SKILLS

Programming: C, C++, Python, Java, PHP
Distributed and Data Systems: Hadoop MapReduce, Hadoop Streaming, HDFS, distributed graph analytics, distributed algorithms
Machine Learning and Scientific Computing: TensorFlow, Keras, PyTorch, OpenCV, NumPy, pandas, NLTK, spaCy
Machine Learning Concepts: CNN, LSTM, RNN, GAN
Web and Visualization: D3.js v7, Laravel, frontend development

AWARDS AND SCHOLARSHIPS

- International Travel Award (USD $1500), Graduate Student Senate, Kent State Univerisity October 2026
- Graduate Research Assistantship, Ph.D. in Computer Science, Kent State University Aug 2024 – Present
- Full Scholarship, Bachelor's in Computer Engineering, Institute of Engineering, Tribhuvan University Nov 2017 – Apr 2022

SERVICES AND VOLUNTEERING

Graduate Student Senator, Department of Computer Science, Kent State University Sept 2026 – Present
Vice-President/Member, Nepalese Student Association at Kent State University Aug 2024 – July 2026
President, Student Electronics and Computer Club Apr 2021 – Apr 2022
Himalaya College of Engineering, Tribhuvan University, Kathmandu, Nepal

SELECTED COURSEWORKS

Graduate Coursework: Design and Analysis of Algorithms; Advanced Topics in Algorithms; Math Foundations on Machine Learning and AI; Computational Geometry; Data Mining Techniques; Big Data Analytics; Information Visualization; Advanced Computer Graphics; Graph and Social Network Analysis; Internet of Things
Undergraduate Coursework: Data Structures and Algorithms; Object-Oriented Programming; Discrete Mathematics; Probability and Statistics; Theory of Computation; Computer Organization and Architecture; Artificial Intelligence; Computer Networks; Computer Graphics; Database Management Systems; Operating Systems; Software Engineering